Yihao Sun

AI Infrastructure / AI Engineering Internship Candidate

yihaosun@andrew.cmu.edu | GitHub: Alexcel-Harry | LinkedIn: Yihao Sun

Phone: +1 (616) 227-9785; +86 13916963098

Availability: May 2026 – August 2026

PROFESSIONAL PROFILE

A graduate student at Carnegie Mellon University with strong foundations in machine learning and systems. Experienced in building, integrating, and evaluating end-to-end AI systems under real-world constraints, including edge deployment, simulation platforms, and performance-critical training pipelines. Also skilled in developing, evaluating, and deploying learning-based models. Seeking for an AI infrastructure, AI Engineering, or applied research internship focused on scalable and reliable AI systems.

EDUCATIONAL BACKGROUND

Carnegie Mellon University — Pittsburgh, PA

Master of Science in Mobile & Internet of Things Engineering — Anticipated May 2027

GPA: 3.85/4.0

University of Michigan — Ann Arbor, MI

Bachelor of Science in Engineering, Major in Data Science, Minor in Mathematics — Aug 2023 - May 2025

GPA: 3.86/4.0

Shanghai Jiao Tong University — Shanghai, China

Bachelor of Engineering, Major in Electrical and Computer Engineering, Minor in Computer Science — Sep 2021 - Aug 2025

GPA: 3.53/4.0

Honors & Awards: Dean's List (2023, 2024), University Honor (2023, 2024, 2025)

Relevant Coursework: Computer Vision, Conversational AI, Database Systems, Distributed Systems, Machine Learning Research Experience, Mobile and Pervasive Computing

In Progress (Spring 2026): Generative AI, Large Language Model Systems, Robotics Learning

RESEARCH EXPERIENCE & PUBLICATIONS

End-to-End Adaptive Smart Glasses for Caregiving — Oct 2025 - Present (Early-staged)

Carnegie Mellon University - Privacy-First Wearable AI Research,

- Developing a privacy-first wearable assistant using an edge Vision-Language Model (VLM) and Multi-Timescale Meta-Continual Learning framework that dynamically adapts sensor fusion between chronic monitoring (long-term health tracking) and acute injury modes (immediate emergency response).
- Implementing a Dual Working Memory system with Reinforcement Learning to personalize intervention timing, optimizing the complex tradeoff between assistance utility and user disruption—balancing real-time constraints with long-term adaptation through continuous learning.
- Designing the system architecture to operate entirely on edge hardware, ensuring data privacy while maintaining responsiveness for time-sensitive caregiving scenarios.

Part-Level Scene Understanding with LLMs — May 2025 – Sept 2025

Open-Source Research Project — `https://github.com/PSGBOT/KAFNet_Pipeline`

- Constructed the comprehensive Part-Segmentation-Relation dataset through an automated pipeline that fuses Semantic-SAM for hierarchical segmentation with Gemini 2.5 for detailed kinematic annotation of part functionalities and relationships.
- Designed and implemented KAF-Net, a novel scene graph generation model that predicts fine-grained part relationships using Kinematic Affinity Fields and Multi-Channel Masks to capture both spatial and functional dependencies.

- Enabled kinematic-aware robotic task planning by integrating KAF-Net with Large Language Models to generate executable action sequences via dynamic graph pruning, integrating the SayPlan framework for practical robotic manipulation.
- **Publication Status:** *Co-first Author* manuscript titled *"KAF-Net: Towards Part-Informed Scene Graphs for Fine-Grained Robotic Manipulation"* currently under revision for RA-L.

TeraSim: Generative Simulation for Autonomous Vehicle Safety Mar 2024 – Apr 2025
MCITY Lab Research — https://arxiv.org/abs/2503.03629

- Engineered a novel co-simulation plugin that integrates SUMO (traffic flow simulation) and CARLA (3D environment simulation) to inject dynamic roadblocks, vulnerable road users, and agent misfeasance into autonomous vehicle testing loops.
- Converted and refined OpenDrive maps for high-fidelity simulation of the MCITY test environment, improving map accuracy and providing critical feedback to the facility team for physical test track enhancements.
- **Publication Status:** Co-authored paper *"TeraSim: Uncovering Unknown Unsafe Events for Autonomous Vehicles through Generative Simulation"* submitted to IROS 2025.
- This research serves as the technical foundation for **SaferDrive.ai** (https://www.saferdrive.ai/), a startup focused on generative AI simulation for autonomous vehicle safety.

Magnetic Wristband Gesture Recognition Mar 2022 – Mar 2023
Shanghai Jiao Tong University Undergraduate Research

- Optimized hardware configuration and data processing pipelines to achieve >80% accuracy in full-finger gesture classification with high noise resistance in real-world environments.
- Developed noise-resistant signal processing techniques to mitigate external magnetic interference, implementing advanced filtering algorithms to isolate gesture signals from environmental noise.
- Mapped gestures via Euclidean space geometric center analysis, creating a robust classification system that maintains accuracy across different users and environmental conditions.

TECHNICAL SKILLS & PROFICIENCIES

Programming Languages: Python, C/C++, CUDA, SQL, R, Go, System Verilog, Java

AI & Machine Learning Tools: PyTorch, TensorFlowLite, Scikit-learn, NumPy, Pandas, Matplotlib, stable_baselines3, OpenCV, CARLA, SUMO, MuJoCo

Systems & Development Tools: Git, Docker, Linux/Unix, Web Systems, Databases

Computer Architecture & Hardware: RISC-V ISA, ARM ISA, Branch Prediction Algorithms, Superscalar CPU Pipelines

Concepts & Methodologies: Databases, Data Mining, Distributed Systems, Information Security, LLM Distillation, LLM Finetuning, Model Quantization, Reinforcement Learning, Robotics Learning,

Languages: English (Native/Bilingual Proficiency; TOEFL 112/120), Chinese (Native)

TECHNICAL PROJECTS & IMPLEMENTATIONS

Real-time Multiple Object Tracking on Edge Hardware Sept 2025 - Dec 2025
Mobile and Pervasive Computing Course Project

- Engineered a real-time Multiple Object Tracking pipeline on Google Pixel 7/9 Pro smartphones, utilizing on-device TPU acceleration for YOLO11 and ByteTrack for multi-object tracking.
- Implemented optical-flow-based prediction for intermediate frames to ensure stable and smooth 30 FPS tracking while maintaining accuracy.
- Developed techniques for keyframe ratio optimization and adaptive frame dropping to handle varying computational loads while preserving tracking continuity.

Impossible Distillation & LoRA Fine-Tuning Sept 2024 - Dec 2024
Machine Learning Research Experience Capstone Project

- Reproduced the "Impossible Distillation" framework to generate high-quality synthetic paraphrase datasets from GPT-2, implementing the complete dataset generation pipeline with rigorous semantic and syntactic evaluation metrics.
- Integrated Low-Rank Adaptation with the distillation pipeline, achieving a >99% reduction in trainable parameters while slightly improving model performance on paraphrasing tasks.
- Designed and implemented the parameter-efficient fine-tuning process, demonstrating practical approaches for adapting large language models with limited computational resources.
- Validated the pipeline quality through comprehensive evaluation including BLEU scores, semantic similarity metrics, and human evaluation studies.

3-Way Superscalar RISC-V Processor May 2025 - Aug 2025
Computer Architecture Capstone Project

- Designed, implemented, and optimized a high-performance, 3-way superscalar RISC-V processor from the ground up in System Verilog, implementing a 5-stage pipeline with comprehensive data and control hazard handling.
- Enhanced instruction throughput by designing an advanced TAGE branch predictor with aging useful bits mechanism, reducing misprediction rates to achieve a 30% performance uplift compared to baseline predictors.
- Developed extensive verification testbenches and performance analysis tools to validate correctness and measure architectural improvements under various workload conditions.

Distributed Sharded Key-Value Store Series Jan 2025 – Apr 2025
Distributed Systems Course Project — Go, Paxos, Distributed Consensus

- Architected a high-performance distributed storage system, evolving it from a Primary/Backup replication model managed by a central View Service to a scalable, Sharded Multi-Paxos architecture.
- Implemented a robust Paxos consensus library to manage replicated state machines, ensuring linearizable consistency and partition tolerance without relying on a central coordinator.
- Designed a ShardMaster configuration service to partition data across replica groups, employing a deterministic load-balancing algorithm to dynamically rebalance shards during cluster scaling with minimal data movement.
- Engineered fault-tolerance mechanisms including a View Service for automatic failover in the initial design, and later implemented log compaction and concurrent RPC handling to optimize the Paxos-based implementation.

Fake News Detection System Sep 2024 - Dec 2024
Conversational AI Course Project

- Architected a Retrieval-Augmented Generation based verification system that cross-references user queries against a curated database of credible sources, providing evidence-based fact-checking responses.
- Engineered a backend data pipeline to automatically crawl, process, and index news articles from reputable sources, enabling efficient, low-latency information retrieval with semantic search capabilities.
- Designed the system with scalability in mind, supporting real-time updates to the knowledge base as new information becomes available.

No-Limit Texas Hold'em AI Agent Jan 2026 - Present
Personal Research Project — https://github.com/Alexcel-Harry/poker_bot

- Developed a Deep Counterfactual Regret Minimization (Deep CFR) agent for 3-player No-Limit Hold'em, utilizing separate advantage and strategy networks to approximate Nash Equilibrium.
- Engineered a high-throughput training pipeline with 64-way parallelized environment traversals and batched FP16 inference, reducing convergence time to < 8 hours.
- Designing an action abstraction layer to map continuous betting dynamics into discrete strategy bins, overcoming the sparsity issues of No-Limit action spaces.